Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/11/2015
PI Name:	Nielsen, Sheila Ph.D.		
Project Title:	Genotypic and phenotypic responses of Candida albicans to spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	PI name change to Sheila Nielsen in 2014 (formerly Sheila Nielsen-Preiss)--Ed., 1/12/2015		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	04/01/2012	**End Date:**	07/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	ISS		
Flight Assignment:	Space-X ; ISS NOTE: Extended to 7/31/2017 per NSSC information (Ed., 7/13/16) NOTE: Extended to 7/31/2016 per A. Chu/ARC and NSSC information (Ed., 8/12/15) NOTE: Extended to 8/31/2015 for reflight as Micro-8 experiment, per A. Chu/ARC and NSSC information (Ed., 4/3/14) NOTE: Extended to 8/31/2014 per A. Chu/ARC and NSSC information (Ed., 6/17/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12AH29G		
Performance Goal No.:			
Performance Goal Text:			

	Micro-8 Experiment (Addendum March 2014 to "Genotypic and phenotypic responses of Candida albicans to spaceflight") We had the opportunity to fly the Micro-6 payload of C. albicans experiments on SpX CRS-1 in October 2012 and many of the experimental results from this study were positive and intriguing. Subsequent to this initial flight experience, we are in a position to repeat several of the experiments to validate the data and provide power to the analyses. In addition, pursuant to the lessons learned, internally by our team and more generally as part of the initial Space Exploration expedition, we feel refining the experimental design and altering the sample handling processes will strengthen the data. The goal of our ongoing studies is to further examine the responses of C. albicans to microgravity by performing hypothesis-driven research in spaceflight conditions. Our overarching hypothesis is that exposure of C. albicans to microgravity will alter gene expression and morphology, consistent with a potential increase in virulence. Notably, these studies 1) will inform the value of LSMMG for predicting the physiological responses of C. albicans, 2) will further explore and document the phenotypic parameters of C. albicans that are associated with pathogenicity and are altered during exposure to spaceflight, and 3) may predict conserved responses of higher eukaryotes, including humans, to spaceflight conditions. This re-flight opportunity of the C. albicans payload will strengthen and extend the data as well as focus the analysis and interpretation. (Ed. note 4/8/14: reporting for the Micro-8 addendum will begin in July 2014)
Task Description:	Micro-6 Experiment (Addendum to: "The fungal response to simulated microgravity, grant NNX09AM79G") Modeling the microgravity environment, using rotating suspension culture bioreactors, has allowed investigation of eukaryotic cellular responses in a ground-based model and has provided insights into the impact of spaceflight on cellular physiology. Extrapolation of results from modeled systems to that expected in space is useful and informative, but evaluating the impact of actual space exposure strengthens the validity and interpretation of data from both environments. Using the Candida albicans (C. albicans) yeast system, we have demonstrated that simulated microgravity affects differential expression of several groups of genes, including those related to morphogenic transformation, environmental sensing and ergosterol biosynthesis; induces morphologic changes such as filamentation, biofilm formation and colony morphology; and results in an increased resistance to the antimicrobial agent Amphotericin B. The constellation of these physiological responses is consistent with a potential increase in the virulence of this opportunistic yeast pathogen. The focus of the current proposal is to characterize physiological outcome measures of C. albicans following exposure to flight on the International Space Station (ISS). The overall objective of this proposal is to evaluate the response of C. albicans to spaceflight in order to 1) predict the potential health risk posed by this opportunistic pathogen in the spaceflight environment, and 2) to further characterize the phenotypic and genotypic response of a eukaryotic system to conditions of microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The microgravity environment is also one of low fluid shear. Some of our unpublished data suggests that this low fluid shear environment may be contributing significantly to the yeast response. Although there are clear implications for yeast grown in space, there are also microenvironments within the body that might be classified as low fluid shear, such as the brush border regions in the digestive tract. We hope to be able to extrapolate the results from the low shear flight environment to the response of Candida albicans in the human digestive system.
Task Progress:	This was an exciting year of opportunity. Following a presentation of our research and several discussions at the 2nd annual ISS Research and Development Conference in Denver, CO, July 2013, we pursued funding for a re-flight opportunity. The proposal addendum was recommended for funding in late February 2014 and we began the process of developing the Micro-8 payload, scheduled to launch on SpX-4. This preparation process included continued analysis of the Micro-6 experimental results, testing parameters for the Micro-8 experiments, and training new team members since four of our team completed their studies at Montana State University (as either graduate or undergraduate students) and were accepted to medical school (3) or began a teaching experience (1). The goal of the ongoing studies continues to be to further examine the responses of C. albicans to microgravity by performing hypothesis-driven research in spaceflight conditions. Our overriding hypothesis is that exposure of C. albicans to microgravity will alter gene expression and morphology, consistent with a potential increase in virulence. Notably, these studies 1) will inform the value of LSMMG for predicting the physiological responses of C. albicans, 2) will further explore and document the phenotypic parameters of C. albicans that are associated with pathogenicity and are altered during exposure to spaceflight, and 3) may predict conserved responses of higher eukaryotes, including humans, to spaceflight conditions. Modeling the microgravity environment, using rotating suspension culture bioreactors, has allowed investigation of eukaryotic cellular responses in a ground-based model and has provided insights into the impact of spaceflight on cellular physiology. Extrapolation of results from modeled systems to that expected in space is useful and informative, but evaluating the impact of actual space exposure strengthens the validity and interpretation of data from both environments. Using the Candida albicans (C. albicans) yeast system, we have demonstrated that simulated microgravity affects differential expression of several groups of genes, including those related to morphogenic transformation, environmental sensing, and ergosterol biosynthesis; induces morphologic changes such as filamentation, biofilm formation, and colony morphology; and results in an increased resistance to the antimicrobial agent Amphotericin B. The constellation of these physiological responses is consistent with a potential increase in the virulence of this opportunistic yeast pathogen. A more complete understanding of the yeast adaptation response to extreme environments and the potentially associated health risks is vital for long-term flight crew health and safety.
Bibliography Type:	Description: (Last Updated: 06/23/2023)
Significant Media Coverage	NASA TV interview with Lori Meggs. "The current status of NASA-supported research at the ISS National Laboratory; PI Nielsen interviewed." NASA TV interview with Lori Meggs. Taped June 2013, Aired January 2014., Jan-2014